HOMEWORK 6.12 · UNIT 6 · NC.8.EE.8 · NC.8.F.4 · NC.8.F.3 · NC.8.F.2

Systems of Equations (Graphing)

Targets: EE.8-2, EE.8-3, EE.8-1, F.4-4, F.3-4, EE.8-5, EE.8-6, F.2-4, EE.8-8, EE.8-9

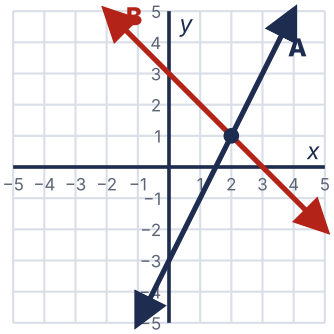
Name _____ Date _____ Block _____

Remember: The **solution** of a system is the point where the lines **cross**. It makes **both** equations true. Different slopes → **one** solution. Same slope, different b → **no** solution (parallel). Same slope, same b → **infinitely many**. **Substitution:** if $y = 2x + 1$ and $y = -x + 7$, set $2x + 1 = -x + 7$ → $x = 2$, $y = 5$.

SCORE 3.0 Approaching · the basics

Targets: EE.8-2, EE.8-3, EE.8-1, F.4-4, F.3-4

1 Line A is $y = 2x - 3$. Line B is $y = -x + 3$. What is the solution? EE.8-2



2 How many solutions? $y = -x + 4$ and $y = -x + 1$ EE.8-3

Ⓐ One
Ⓑ None
Ⓒ Infinitely many

3 Is (2, 5) a solution of $y = 3x - 1$ and $y = x + 2$? EE.8-1

Yes No

Why? _____

4 Write $3x + y = 4$ in slope-intercept form. REVIEW 6.10 F.4-4

5 The x-intercept of $2x + 3y = 6$ is (3, 0). REVIEW 6.11 F.3-4

True False

SCORE 4.0 Proficient · on target Targets: EE.8-5, EE.8-6, F.2-4

6 Solve by substitution: $y = x + 5$ and $y = 3x - 1$ EE.8-5

7 **Priya** was asked how many solutions $y = 4x + 1$ and $y = -4x + 1$ have. Find the mistake. Fix it. EE.8-6

PRIYA'S WORK
Both equations have a 4 and a 1.
So the lines are parallel.
No solution.

8 A is $y = 6x + 2$. B is the table. Which is true? REVIEW 6.9 F.2-4

Function B

x	0	1	2
y	5	9	13

(A) A grows faster. B starts higher.
(B) B grows faster. B starts higher.
(C) A grows faster. A starts higher.
(D) They grow at the same rate.

SCORE 5.0 Mastery · go deeper Targets: EE.8-8, EE.8-9

9 Movie Club A charges \$8 per movie: $y = 8x$. Club B charges a \$20 fee plus \$4 per movie: $y = 4x + 20$. Solve the system. When is Club B cheaper? EE.8-8

The solution is _____ . This means _____ . Club B is cheaper when _____ .

10 How many solutions do $y = 3x + 2$ and $6x - 2y = -4$ have? Explain using slopes and y-intercepts. EE.8-9

In slope-intercept form, the second equation is _____ . The slopes are _____ and the y-intercepts are _____ . So _____ .